

JEE MAIN CRASH COURSE

IT'S TIME TO LEARN APPLICATION

- ✓ LATEST NTA PATTERN
- ✓ CONCEPT VIDEOS
- ✓ LIVE DOUBT SESSIONS
- ✓ DPP, REVISION SHEETS
- ✓ 100% MONEYBACK TO MERITORIOUS STUDENTS

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Designed by
Math Maestro

Anup Sir



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JEE Mains 2020 Jan Chapter wise Question Bank

Quadratic Equation

01

Let α and β are the roots of $x^2 - x - 1 = 0$ such that $P_k = \alpha^k + \beta^k$, $k \geq 1$ then which one is incorrect?
माना कि α तथा β समीकरण $x^2 - x - 1 = 0$ के मूल इस प्रकार है कि $P_k = \alpha^k + \beta^k$, $k \geq 1$ तो निम्न में से कौनसा गलत है—

- (1) $P_5 = P_2 \times P_3$
(3) $P_3 = P_5 - P_4$

- (2) $P_1 + P_2 + P_3 + P_4 + P_5 = 26$
(4) $P_4 = 11$

7th Jan Evening

Sol

(1)

$$\alpha^5 = 5\alpha + 3$$

$$\beta^5 = 5\beta + 3$$

$$P_5 = 5(\alpha + \beta) + 6$$

$$= 5(1) + 6$$

$$P_5 = 11 \text{ and तथा } P_5 = \alpha^2 + \beta^2 = \alpha + 1 + \beta + 1$$

$$P_2 = 3 \text{ and तथा } P_3 = \alpha^3 + \beta^3 = 2\alpha + 1 + 2\beta + 1 = 2(1) + 2 = 4$$

$$P_2 \times P_3 = 12 \text{ and तथा } P_5 = 11 \Rightarrow P_5 \neq P_2 \times P_3$$

02

If $2x^2 + (a - 10)x + \frac{33}{2} = 2a$, $a \in \mathbb{Z}^+$ has real roots, then minimum value of 'a' is equal to

8th Jan Morning

Sol

8

$$D \geq 0$$

$$(a - 10)^2 - 4(2) \left(\frac{33}{2} - 2a \right) \geq 0$$

$$(a - 10)^2 - 4(33 - 4a) \geq 0$$

$$a^2 - 4a - 32 \geq 0 \Rightarrow a \in (-\infty, -4] \cup [8, \infty)$$

03

Quadratic Equation

JEE Mains 2020 Jan Chapter wise Question Bank

Solution set of $3^x(3^x - 1) + 2 = |3^x - 1| + |3^x - 2|$ contains

- (1) singleton set (2) two elements
(3) at least four elements (4) infinite elements

8th Jan Evening

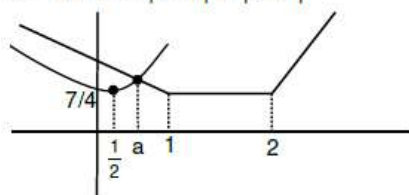
Sol

(1)

Let $3^x = t$

$$t(t-1) + 2 = |t-1| + |t-2|$$

$$t^2 - t + 2 = |t-1| + |t-2|$$



are positive solution

$$t = a$$

$$3^x = a$$

$$x = \log_3 a \text{ so singleton set}$$

04

Find number of real roots of equation $e^{4x} + e^{3x} - 4e^{2x} + e^x + 1 = 0$ is

समीकरण $e^{4x} + e^{3x} - 4e^{2x} + e^x + 1 = 0$ के वास्तविक हलों की संख्या है—

- (1) 1 (2) 2 (3) 3 (4) 4

9th Jan Morning

Sol

Quadratic Equation

JEE Mains 2020 Jan Chapter wise Question Bank

(1)

Let $e^x = t \in (0, \infty)$

Given equation दी गई समीकरण

$$t^4 + t^3 - 4t^2 + t + 1 = 0$$

$$t^2 + t - 4 + \frac{1}{t} + \frac{1}{t^2} = 0$$

$$\left(t^2 + \frac{1}{t^2}\right) + \left(t + \frac{1}{t}\right) - 4 = 0$$

$$\text{Let } t + \frac{1}{t} = \alpha$$

$$(\alpha^2 - 2) + \alpha - 4 = 0$$

$$\alpha^2 + \alpha - 6 = 0$$

$$\alpha^2 + \alpha - 6 = 0$$

$$\alpha = -3, 2 \Rightarrow \alpha = 2 \Rightarrow e^x + e^{-x} = 2$$

$x = 0$ only solution

05

Let both root of equation $ax^2 - 2bx + 5 = 0$ are α and root of equation $x^2 - 2bx - 10 = 0$ are α and β .

Find the value of $\alpha^2 + \beta^2$

माना समीकरण $ax^2 - 2bx + 5 = 0$ के दोनों मूल α हैं एवं समीकरण $x^2 - 2bx - 10 = 0$ के मूल α तथा β हैं, तब $\alpha^2 + \beta^2$ का मान ज्ञात करो।

(1) 20

(2) 25

(3) 15

(4) 30

9th Jan Evening

Sol

(2)

$$2\alpha = \frac{2b}{a} \Rightarrow \alpha = \frac{b}{a} \text{ and } \alpha^2 = \frac{5}{a} \Rightarrow \frac{b^2}{a^2} = \frac{5}{a}$$

$$\Rightarrow b^2 = 5a \dots\dots (i) \quad (a \neq 0)$$

$$\alpha + \beta = 2b \dots\dots\dots(ii)$$

$$\text{and तथा } \alpha\beta = -10 \dots\dots\dots(iii)$$

$$\alpha = \frac{b}{a} \text{ is also root of } x^2 - 2bx - 10 = 0$$

$$\alpha = \frac{b}{a}, x^2 - 2bx - 10 = 0 \text{ का मूल है}$$

$$\Rightarrow b^2 - 2ab^2 - 10a^2 = 0$$

$$\text{by (i) से } \Rightarrow 5a - 10a^2 - 10a^2 = 0$$

$$\Rightarrow 20a^2 = 5a$$

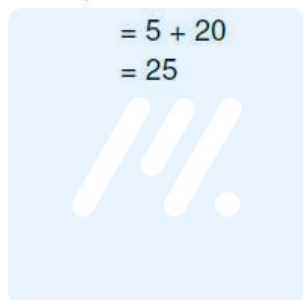
$$\Rightarrow a = \frac{1}{4} \text{ and तथा } b^2 = \frac{5}{4}$$

$$\alpha^2 = 20 \text{ and तथा } \beta^2 = 5$$

Now अब $\alpha^2 + \beta^2$

$$= 5 + 20$$

$$= 25$$



mathongo

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